# Introduction to Digital Forensics

## Start to Present

## What, Why, Where, Who, How?

**NSSECU3 | Digital Forensics**

| | |
|---|---|
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# Overview

1. What is Digital Forensics?
2. Goal of Forensics Analysis
3. Digital Forensics Process*
4. Where is Digital Forensics Involved?
5. Uses of Digital Forensics Analysis
6. Incident Response Life Cycle*
7. Digital Forensics Core Methodology
8. Investigative Process

# What is Digital Forensics?

A review and an overview

# Digital Forensics

- The underlying methodology that gives examiners the <u>ability to answer</u> the **who**, **what**, **why**, **when**, **where**, and **how** questions.

SANS DFIR

DIGITAL FORENSICS & INCIDENT RESPONSE

# Art and Science

## Art

Art of Investigation

Interpreting data to reconstruct an event

Investigative Approach

Combine
With

## Science

Science of Investigation

Applying the scientific method and deductive reasoning

Iterative Approach

# Goal of Forensic Analysis

Just Facts

# Goal of Forensic Analysis

**Find Facts** → **Via these facts, recreate an event**

The examiner reveals the truth of an event by discovering and exposing the **remnants** of the event that have been left on the system.

# Goal of Forensic Analysis

Examiner reveals the truth of an event by discovering and exposing the **remnants** of the event that have been left on the system.

## *Evidence*

Used during legal proceedings. ***Using this term loosely may get an examiner into trouble***

## *Artifact*

***Traces left behind*** due to activities and events, which can be innocuous or not

# Goal of Forensic Analysis

## Every contact leaves a trace.

Even the act of cleaning up artifacts can leave additional artifacts or digital footprint—the digital equivalent to the smell of bleach at a physical crime scene that has been "washed"

# Goal of Forensic Analysis (Motto)

| | |
|---|---|
| Prepare for trouble! | And make it double! |
| To protect the world from devastation! | To unite all peoples within our nation! |
| To denounce the evils of truth and love! | To extend our reach to the stars above! |
| Jessie! | James! |
| Team Rocket blasts off at the speed of light! | Surrender now, or prepare to fight! |
| | |
| Meowth! | That's right! |
| | |

# Goal of Forensic Analysis (Real Motto)

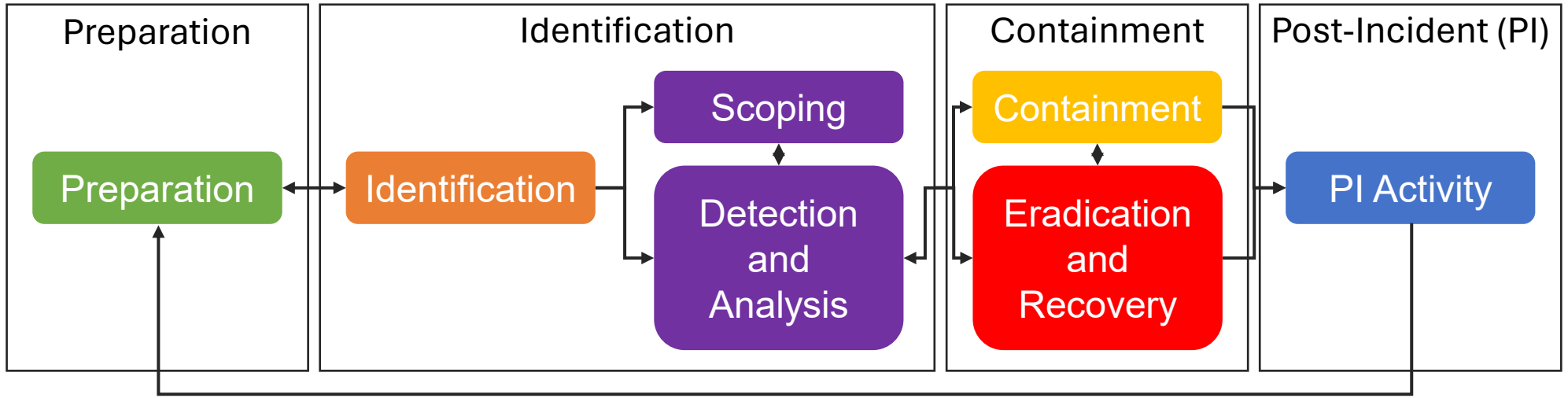
| | |
|---|---|
| Examiner's Goal: | To determine the TRUTH |
| The examiner should start with a hypothesis: | "This person is the one sending hate mails to all of us" |
| The examiner's task is | **NOT** to prove these assertions |
| The examiner's task is | To **uncover** artifacts that indicate the hypothesis to be either valid or invalid. |

# Digital Forensics Process

Digital Forensics and Incident Response (DFIR) Processes

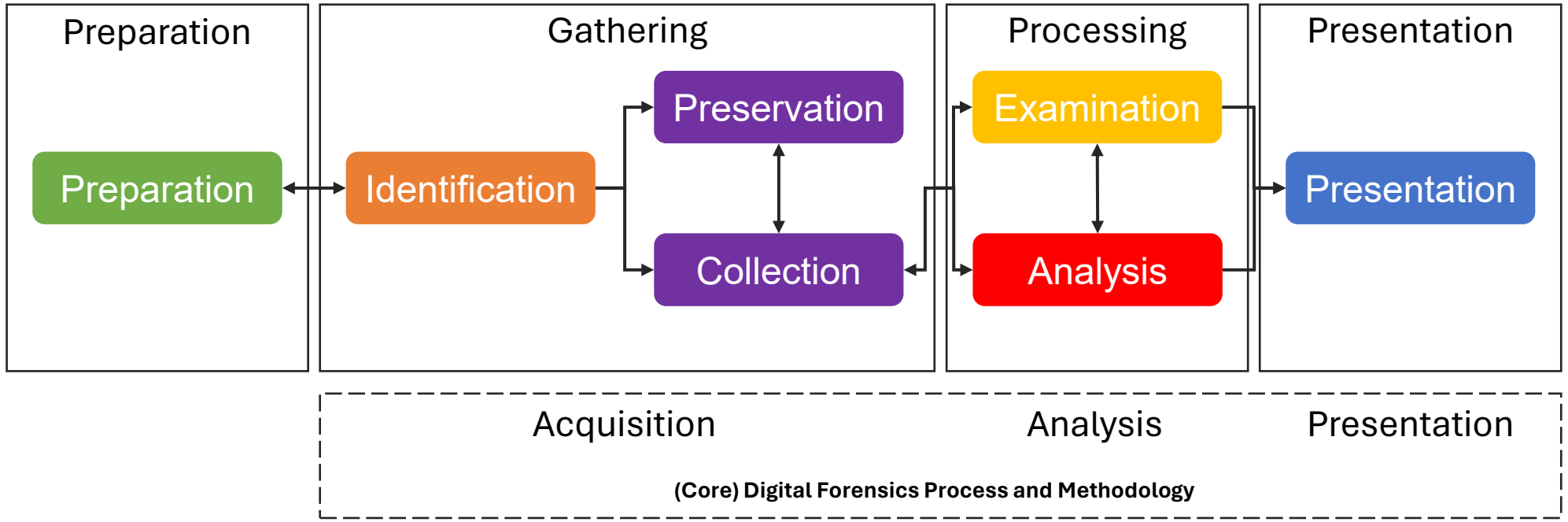
# Incident Response Process (Detailed)

- Set of activities that plan for, detect, and correct the effect of an attack that threatened the security of information



# Digital Forensics Process (Detailed)

- A model as the basis for showing consistent, repeatable, and defensible digital forensic process.



# Digital Forensics Process

Acquisition → Analysis → Presentation

# Digital Forensics Process

## Acquisition

- **Collecting** digital media to be examined
- **Creating** a duplicate of the original media
- **Maintaining** good records of all actions taken with any original media

## Analysis

- Actual media **examination**
- **Identification**, **analysis**, and **interpretation** of artifacts

## Presentation

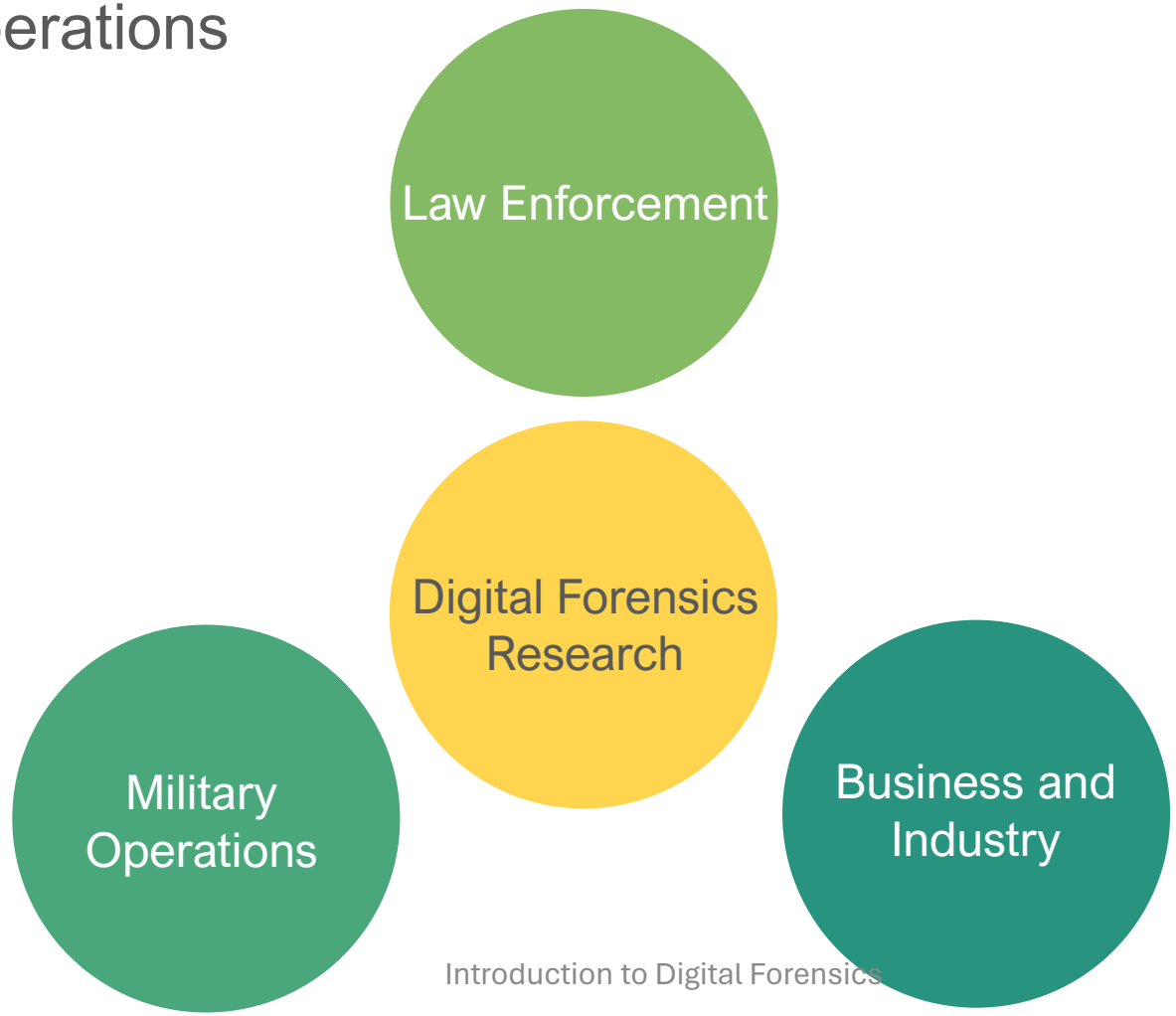
- **Sharing** of results of the analysis phase
- Generating **reports**

# Digital Forensics Involvement

Where is Digital Forensics Involved?

# Where is Digital Forensics Involved?

1. Law Enforcement, Business and Industry, and Military Operations



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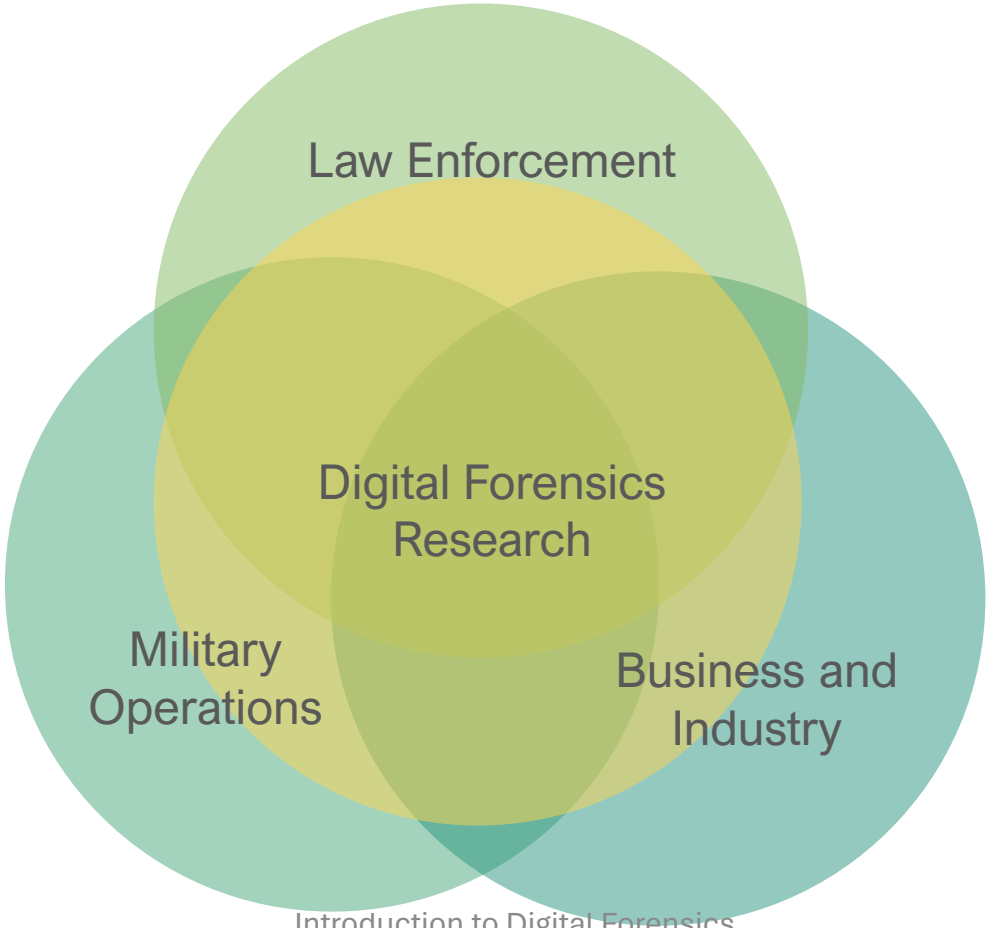
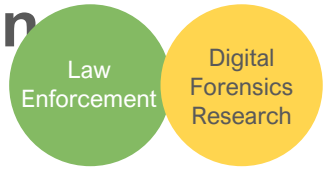


# Where is Digital Forensics Involved?

1. Law Enforcement, Business and Industry, and Military Operations
2. Law Enforcement + DFR = **Court and Litigation**




Business and Industry + DFR = **Critical Infrastructure**

Military Operations + DFR = **Information Warfare**

# Uses of Digital Forensics Analysis

AADDSI

# Uses of Digital Forensics Analysis

- Attribution
- Assessing alibis and statements
- Determining Intent
- Digital Document Verification
- Source evaluation
- **Intrusion Investigation**

A A D D S I

# Uses of Digital Forensics Analysis

- **Intrusion Investigation**

*Why investigate intrusions?*

- Intrusions present large losses to victim organizations
  - Response to **incident**
  - Business **disruption**
  - Damage to **reputation**
  - Lawsuits from **customers / shareholders**
  - Fines for noncompliance with **legal obligations for security**

# Uses of Digital Forensics Analysis

- **Intrusion Investigation**

*Why investigate intrusions?*

Forensics is highly involved in incident response

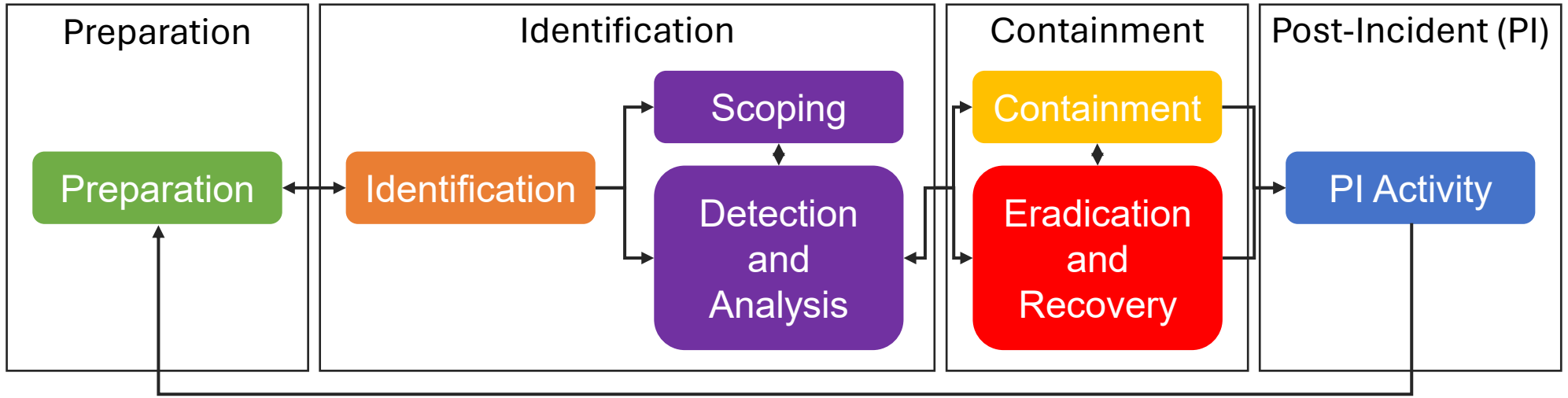
- Goals:
  - To locate and remediate compromised systems
  - Determine if the attacker managed to exfiltrate sensitive information
  - Prove due diligence
  - Apprehend and prosecute the attacker*

# Incident Response Life Cycle

IRLC Again?

# Incident Response Process (Detailed)

- Set of activities that plan for, detect, and correct the effect of an attack that threatened the security of information



# Preparation

Ensuring that the organization (both infrastructure and people) are ready to handle incidents

**Scope:**

- Asset and Data Inventory
- Policies and Procedures
- Host Preparation (e.g. logging and synchronization)
- Logging Infrastructure
- Management Trainings
- Technical Trainings
- The Forensics Toolkit - The Hardware (Jump Bag)

# Preparation

**The Forensics Toolkit - The Hardware (Jump Bag)**

- Laptops - Usually 2 (Analysis and Documentation)
- Blank Removable Media
- For transferring files, storing data etc.
- Loaded Removable Media
- Tools for live acquisition
- Cables (Charging, Networking, etc.)

# Identification / Detection

- Security incidents start in a variety of ways

- Usual indicators:
  - Antivirus alerts
  - IDS alerts
  - Email with suspicious content
  - System and application crashes
  - Blacklist violations
  - Network traffic abnormalities
  - External (third-party) notifications

# Identification

- *How is Digital Forensics Involved in the Identification Phase?*

- Identify the full scope of an intrusion (**Scope assessment**)
  - Actual scope — where intruder activity is verified to have taken place
  - Potential scope — devices that the intruder has access to and may have been compromised as well
- Determine what exactly happened (**Crime Reconstruction**)

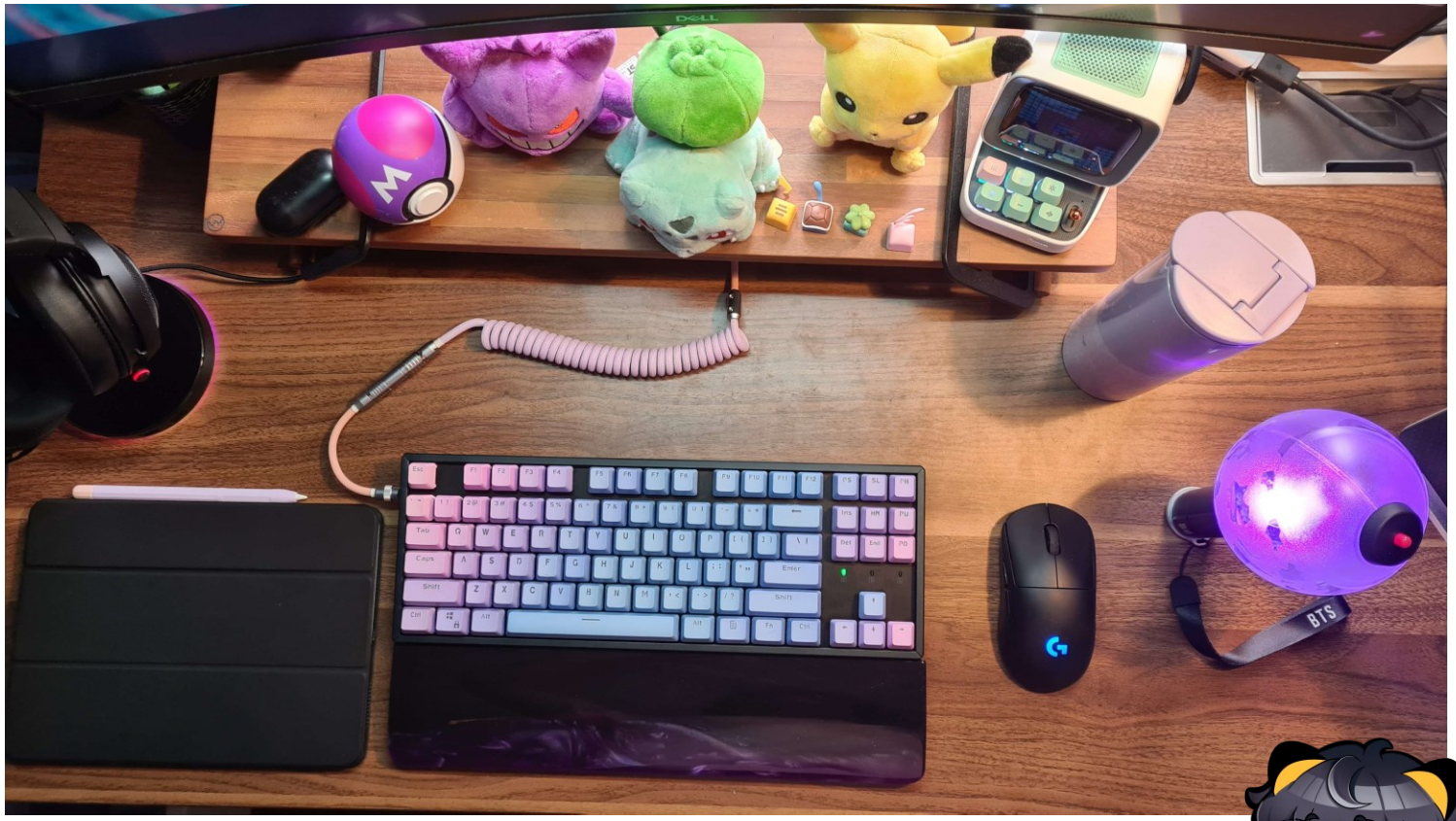
# Scientific Method in IR

Applying the **Scientific Method** in the Context of Incident Response

| | |
|---|---|
| Observation | • Anomalous events indicate an incident and initiate the investigation |
| Hypothesis | • Form a theory of what may have occurred |
| Prediction | • Predict where artifacts may be found |
| Testing | • Look for presence of predicted artifacts and analyze<br>• Test for possible alternative explanations |
| Conclusion | • Form a conclusion based upon results of findings |

# Identification of Electronic Data

Where is the source?



# Identification of Electronic Data

- Typical focus of data acquisition
  - Workstations
  - Mobile Devices
  - Removable Media
  - Servers
  - Backup Environments
  - Network Equipment
  - Image devices (fax, printers, digicams, etc.)

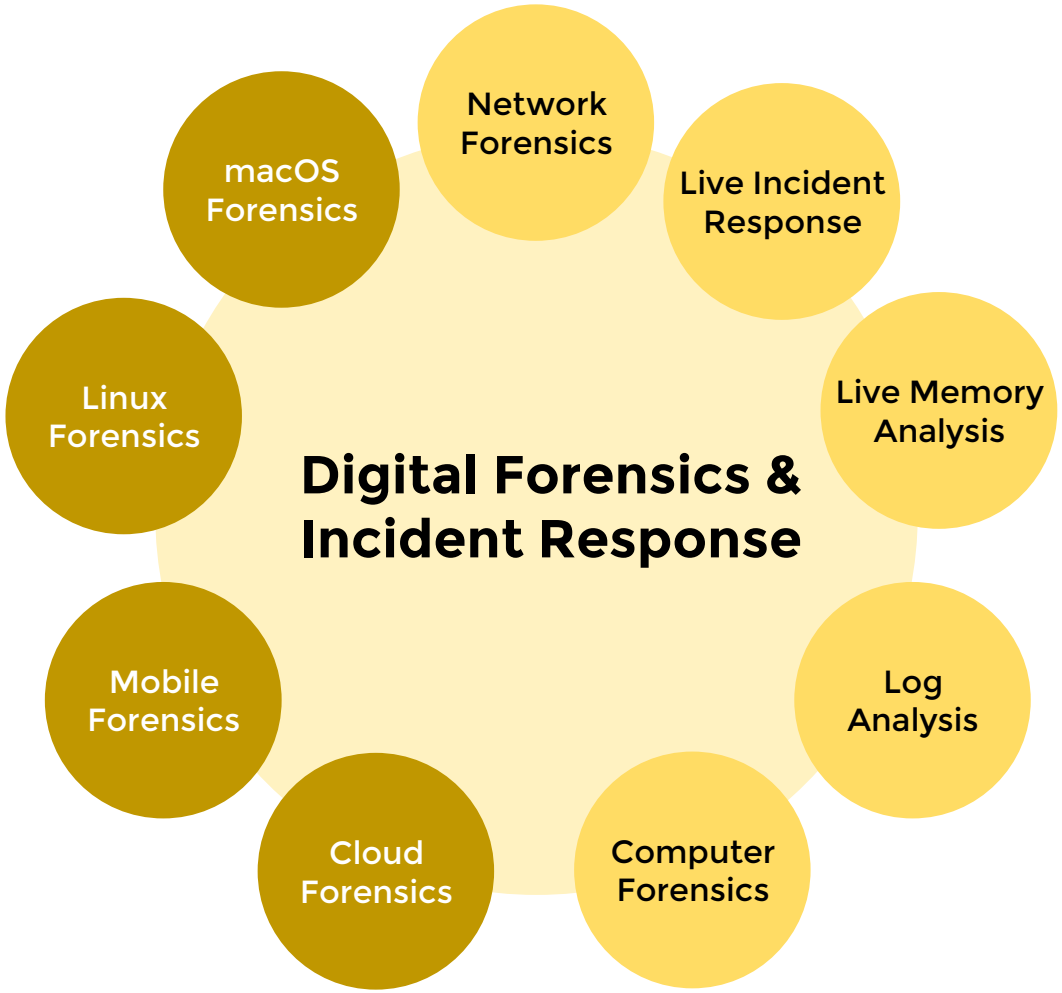
WoRMBINS

# Identification of Electronic Data

- Evidences are everywhere!
  - Computer networks
  - Currently compromised endpoints
  - RAM
  - Logs
  - Hardware
  - Cloud environment
  - Mobile devices
  - Linux OS
  - macOS

# Identification of Electronic Data

- -



# DF Core Methodology

Digital Forensics Core Methodology: Acquisition, Analysis, and Presentation

# Digital Forensics Core Methodology

- **Acquire** artifacts without altering or damaging the origin
  - **Authenticate** that your recovered artifact is the same as the originally-seized data
- **Analyze** the data without modifying it
- **Present** the analyzed data

3A P

# Recall: Digital Forensics Process

Acquire, Analyze, and then Present

# DF Investigative Process

Acquisition → Analysis → Presentation

| Acquisition | | | | | |
|---|---|---|---|---|---|
| Time sync | Imaging | Chain of Custody | Preservation | Sampling | Data Recovery |

| Analysis | | | | |
|---|---|---|---|---|
| Hidden Data Extraction | Validation | Pattern Matching | Traceability | Timeline |

| Presentation | | | |
|---|---|---|---|
| Documentation | Countermeasure | Expert Testimony | Mission Impact Statement |

# DF Core Methodology: Acquire

## Data Collection / Acquisition

- Purely in the context of digital forensics, it involves creating a forensic image of the evidence media
- Most important part of this phase is properly documenting all actions done related to the evidence media to maintain its integrity
- Two types of acquisition: Offline and Live

# DF Core Methodology: Acquire

## Data Collection / Acquisition: Forensic Imaging

- Generating an exact duplicate (or a container that holds an exact duplicate) of a source media
- Involves acquiring a complete sector-for-sector, byte-for-byte copy of the original media
- Complete, meaning aside from active data (data that currently resides on the disk, and we can actually see without specialized  tools), we also need to acquire latent data (deleted data and file slack)

# DF Core Methodology: Acquire

Acquisition → Analysis → Presentation

## Data Acquisition Complexity

- Volatile Data
- Latent Data
- Data Abstraction Layers
- Tool Reliability
- Containment vs Evidence Preservation
- Evidence Dynamics

# DF Core Methodology: Acquire

## Data Acquisition Complexity: Order of Volatility

The Internet Engineering Task Force (IETF) has put together a document titled Guidelines for Evidence Collection and Archiving (RFC 3227) that addresses the order of volatility of digital evidence, as follows:

- Registers and cache
- Routing table, ARP cache, process table, kernel statistics, memory (RAM)
- Temporary filesystems
- Disk
- Remote logging and monitoring data
- Physical configuration, network topology
- Archival media

# DF Core Methodology: Acquire

## Data Preservation

Ensure that artifacts are protected against alteration or destruction.

- Taking pictures
- Use of a primary and a working image
- Purging new media
- Use of message digests
- Using write blockers
- Having a chain of custody
  - Documentation that tracks everything that happens to the artifacts

# DF Core Methodology: Analyze

## Analysis

- Should answer the questions:
  - How - Method of intrusion
  - When - Time/s of access / occurrence
  - Where - Source of attack/s
  - What - Compromised systems and data
- When did the incident start?
- What caused the incident?
- What can we do to prevent this from happening again?
- Aims to reconstruct the crime / incident by correlating events that happened at a specific date and time

# DF Core Methodology: Analyze

## Analysis Methods

- Relational
  - Determines systems involved and their interactions and relationships
- Functional
  - Determines how a system works and how it was configured at the time of the incident
- Temporal
  - Creating the timeline of events to know events leading to the crime (synchronization is important!!)

# DF Core Methodology: Analyze

## Examples of Analysis Tasks

| Network Analysis | Media Analysis | Software Analysis |
|---|---|---|
| Communication Analysis | Disk Imaging | Reverse Engineering |
| Log Analysis | MACtime Analysis (Timelining) | Malicious Code Review |
| Path Tracing | Content Analysis | Exploit Review |
| | Slack Space Analysis | |
| | Steganography | |

# DF Core Methodology: Analyze

## Examples of Analysis Tasks DFIR-Style

- Malware-Related Incident
  - Evidences of File/Folder Opening
  - Evidences of Execution
  - Indicators of Compromise (IOC)
  - Identification/Confirmation/Sweep
  - Endpoint Event and/or Network Logs for Log Analysis
  - Full network sweep of all findings
    - To identify other possibly-compromised systems
  - Root Cause Analysis

# DF Core Methodology: Analyze

## Examples of Analysis Tasks DFIR-Style

- Fraud-Related Incident
  - Evidences of Account Usage
  - Evidences of External Device and Browser Usage
  - Evidences of File/Folder Opening
  - Evidences of Deleted Files and File Knowledge
  - etc.

# DF Core Methodology: Present

## Documentation

- Provides the people involved in the investigation with an overview of the intrusion
- Includes all actions and keeps track of key findings related to an incident
  - Case Summary
  - Forensic Acquisition - very important!
  - Findings and Reporting

# DF Core Methodology: Present

## Documentation

- Check Templates in Module 01

# DF Core Methodology: Present

## Presentation

A lot of artifacts exist! But is it enough?

- This is where the art side of digital forensics comes in
  - A lot of artifacts exist
    - With time, we can identify them all
  - We can gather all the facts that we need
  - Online resources are readily-available
    - Any artifact can be just a quick Google search away!

# DF Core Methodology: Present

## Presentation

A lot of artifacts exist! But is it enough?

- How do we organize our evidences and thoughts?
  - ANSWER
    - It depends
      - On your organization's goal
        - Investigation vs Remediation vs Attribution
      - The requirements needed by a higher authority
        - Litigation
        - Compliance and/or other governing bodies

# DF Core Methodology: Present

## Presentation

- Tailor your presentation based on the ability of your readers
  - No point in showing off technical skills to management
  - No point in explaining business loss and management terms to technical team
- Do not EVER get carried away by bias
  - Stick to the science part
    - Always base your presentation/report on facts
    - Prove/Disprove your hypothesis
    - Leave the ruling to the judging/governing body

# DF Core Methodology: Present

## Presentation

- Read other presentations/documentations/reports
  - Learn from other people
    - Insights on:
      - The way they think
      - Their processes
      - The way they organize their thoughts
      - Tooling and other methodologies they utilized

# DF Core Methodology: Present

## Presentation: Feeding Analysis Back to Detection

- Intrusion artifacts identified can be used to search other systems for similar artifacts
- Identifies additional compromised systems and network segments
- Can predict next targets based on various types of findings (target operating system, subnet, department, etc.)

# That's DFIR in a nutshell!